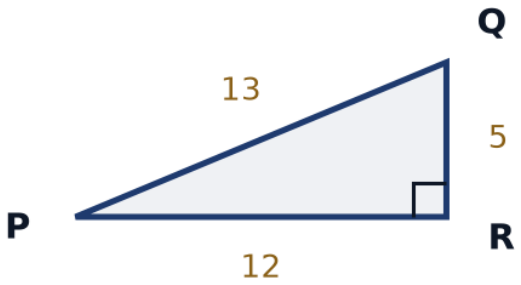
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Solutions to trigonometric equations

For the model $f(x) = 4x + 3$ in solutions to trigonometric equations, evaluate $f(5)$.

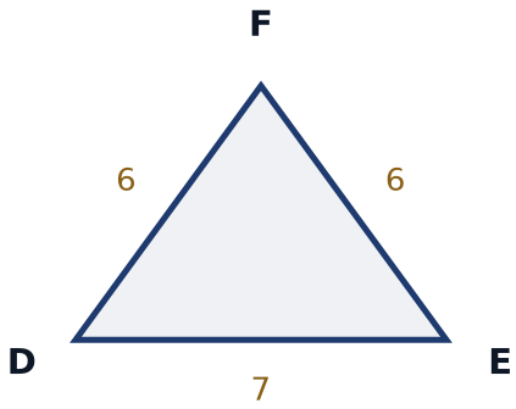


2 [Foundational] • Target: Solutions to trigonometric equations

Find the positive solutions to the quadratic equation $x^2 - 12x + 32 = 0$.

3 [Foundational] • Target: Problem solving with trigonometric functions

For the model $f(x) = 5x + 4$ in problem solving with trigonometric functions, evaluate $f(2)$.



4 [Foundational] • Target: Problem solving with trigonometric functions

Find the positive solutions to the quadratic equation $x^2 - 7x + 6 = 0$.

5 [Foundational] • Target: Problem solving with trigonometric functions

A data set recorded for problem solving with trigonometric functions consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean? Show your complete algebraic work.

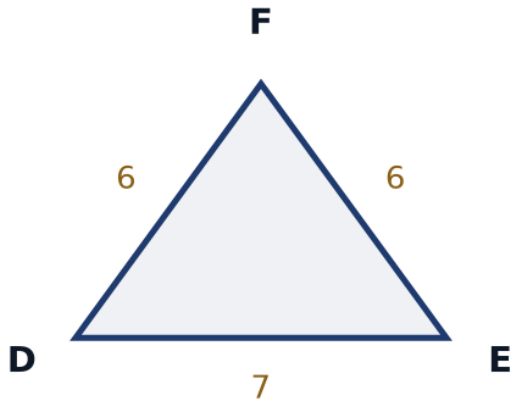
6 [Foundational] • Target: Problem solving with trigonometric functions

For the model $f(x) = 5x + 4$ in problem solving with trigonometric functions, evaluate $f(3)$.

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7 [Foundational] • Target: Problem solving with trigonometric functions

Find the positive solutions to the quadratic equation $x^2 - 9x + 14 = 0$.



Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Problem solving with trigonometric functions

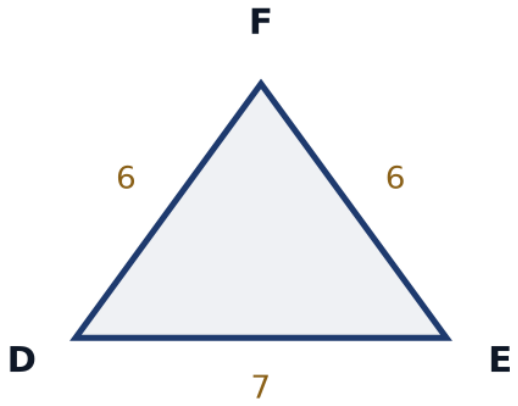
For the model $f(x) = 5x + 4$ in problem solving with trigonometric functions, evaluate $f(4)$.

9 [Intermediate] • Target: Problem solving with trigonometric functions

Find the positive solutions to the quadratic equation $x^2 - 11x + 24 = 0$.

10 [Intermediate] • Target: Problem solving with trigonometric functions

For the model $f(x) = 5x + 4$ in problem solving with trigonometric functions, evaluate $f(5)$.



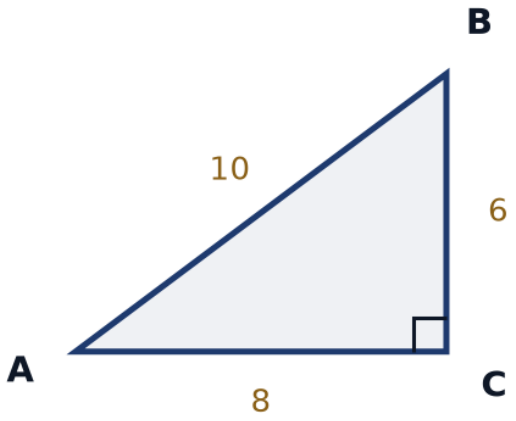
11 [Intermediate] • Target: Problem solving with trigonometric functions

Find the positive solutions to the quadratic equation $x^2 - 13x + 36 = 0$.

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12 [Intermediate] • Target: Double angle identities

In ΔABC studying double angle identities, $m\angle A = 49^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$? Show your complete algebraic work.



13 [Intermediate] • Target: Double angle identities

A geometric figure scaled by factor $k = 2$ has an original side length of 8 cm. What is the corresponding image length? Show your complete algebraic work.

14 [Intermediate] • Target: Double angle identities

On the coordinate plane, a line with slope $m = 2$ passes through $(0, 5)$. What is its y-coordinate when $x = 1$? Show your complete algebraic work.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

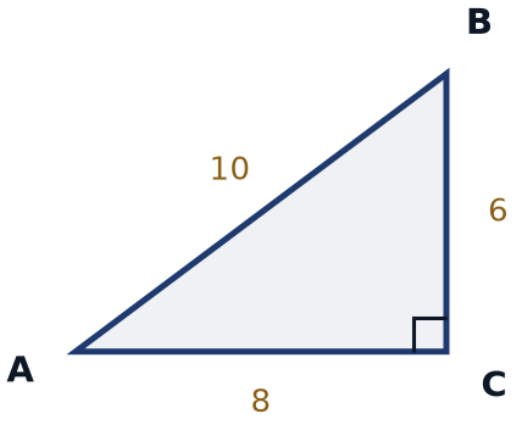
15 [Challenging] • Target: Double angle identities • Cross-Lesson: 5.01 Calculus

In ΔABC studying double angle identities, $m\angle A = 49^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$? Show your complete algebraic work.

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16 [Challenging] • Target: Double angle identities • Cross-Lesson: 1.01 Number & Algebra

A geometric figure scaled by factor $k = 3$ has an original side length of 8 cm. What is the corresponding image length? Show your complete algebraic work.



17 [Challenging] • Target: Double angle identities • Cross-Lesson: 2.01 Functions

On the coordinate plane, a line with slope $m = 2$ passes through $(0, 5)$. What is its y-coordinate when $x = 2$? Show your complete algebraic work.

18 [Challenging] • Target: Double angle identities • Cross-Lesson: 3.01 Geometry & Trigonometry

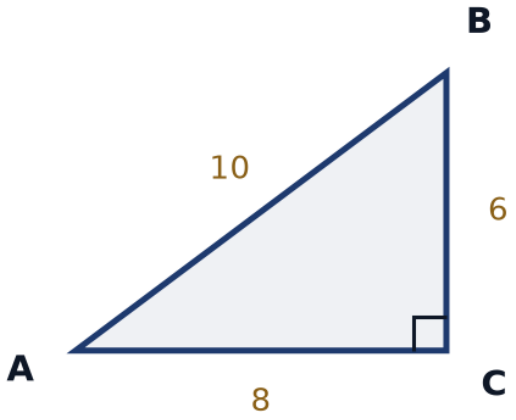
In ΔABC studying double angle identities, $m\angle A = 49^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$? Show your complete algebraic work.

19 [Challenging] • Target: Double angle identities • Cross-Lesson: 4.01 Statistics & Probability

A geometric figure scaled by factor $k = 4$ has an original side length of 8 cm. What is the corresponding image length? Show your complete algebraic work.

20 [Challenging] • Target: Double angle identities • Cross-Lesson: 5.01 Calculus

On the coordinate plane, a line with slope $m = 2$ passes through $(0, 5)$. What is its y-coordinate when $x = 3$? Show your complete algebraic work.

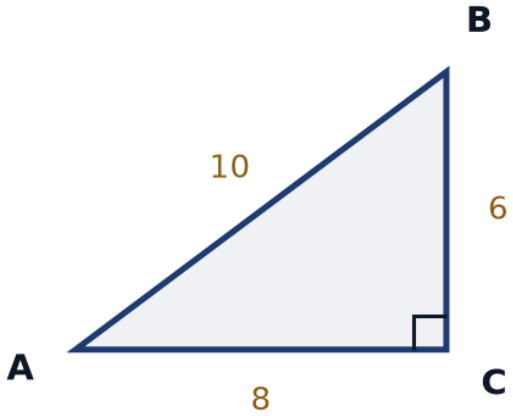


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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Amplitude, vertical shifts and dilations of trigonometric graphs

For the model $f(x) = 2x + 8$ in amplitude, vertical shifts and dilations of trigonometric graphs, evaluate $f(2)$.



a. 12

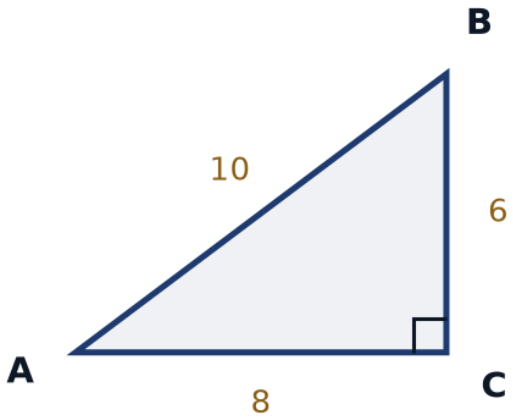
b. 14

c. 4

d. 16

22 [Foundational] • Target: Amplitude, vertical shifts and dilations of trigonometric graphs

Find the positive solutions to the quadratic equation $x^2 - 6x + 8 = 0$.



a. $x = 2$ and $x = 5$

b. $x = 2$ and $x = 4$

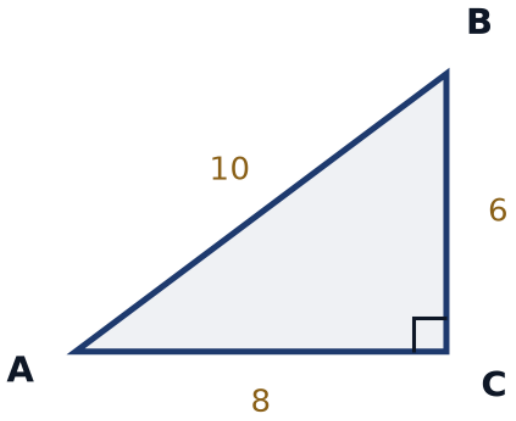
c. $x = -2$ and $x = -4$

d. $x = 6$ and $x = 8$

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23 [Foundational] • Target: Amplitude, vertical shifts and dilations of trigonometric graphs

For the model $f(x) = 2x + 8$ in amplitude, vertical shifts and dilations of trigonometric graphs, evaluate $f(5)$.



a. 18

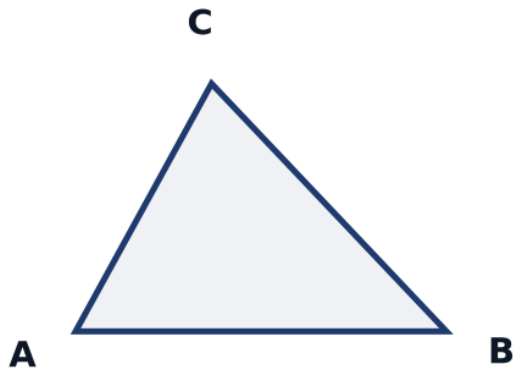
b. 20

c. 10

d. 22

24 [Foundational] • Target: Period and phase shifts of trigonometric graphs

For the model $f(x) = 3x + 9$ in period and phase shifts of trigonometric graphs, evaluate $f(2)$.



a. 15

b. 18

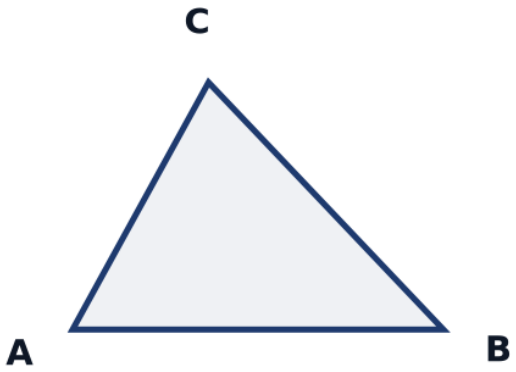
c. 6

d. 19

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25 [Foundational] • Target: Period and phase shifts of trigonometric graphs

Find the positive solutions to the quadratic equation $x^2 - 7x + 10 = 0$.



a. $x = 2$ and $x = 6$

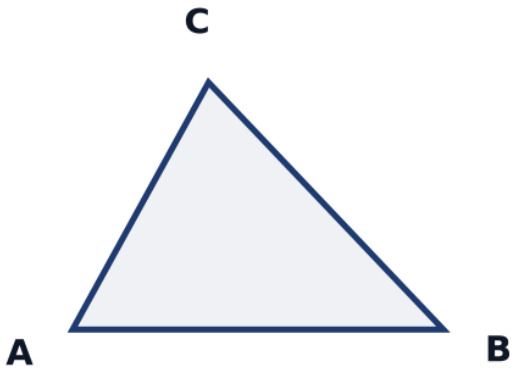
b. $x = 2$ and $x = 5$

c. $x = -2$ and $x = -5$

d. $x = 7$ and $x = 10$

26 [Foundational] • Target: Period and phase shifts of trigonometric graphs

For the model $f(x) = 3x + 9$ in period and phase shifts of trigonometric graphs, evaluate $f(5)$.



a. 24

b. 27

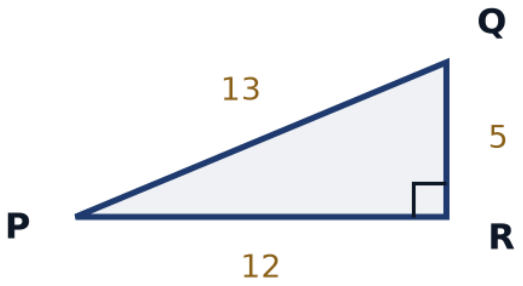
c. 15

d. 28

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27 [Foundational] • Target: Solutions to trigonometric equations

For the model $f(x) = 4x + 3$ in solutions to trigonometric equations, evaluate $f(2)$.



a. 11

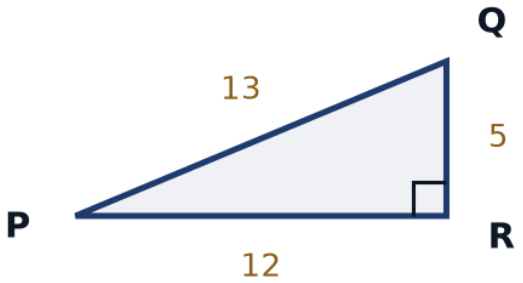
b. 15

c. 8

d. 15

28 [Foundational] • Target: Solutions to trigonometric equations

Find the positive solutions to the quadratic equation $x^2 - 8x + 12 = 0$.



a. $x = 2$ and $x = 7$

b. $x = 2$ and $x = 6$

c. $x = -2$ and $x = -6$

d. $x = 8$ and $x = 12$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Amplitude, vertical shifts and dilations of trigonometric graphs

Find the positive solutions to the quadratic equation $x^2 - 4x + 3 = 0$.

a. $x = 1$ and $x = 4$

b. $x = 1$ and $x = 3$

c. $x = -1$ and $x = -3$

d. $x = 4$ and $x = 3$

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30 [Intermediate] • Target: Amplitude, vertical shifts and dilations of trigonometric graphs

A data set recorded for amplitude, vertical shifts and dilations of trigonometric graphs consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

a. 18.5

b. 19.5

c. 20.5

d. 22.5

31 [Intermediate] • Target: Amplitude, vertical shifts and dilations of trigonometric graphs

For the model $f(x) = 2x + 8$ in amplitude, vertical shifts and dilations of trigonometric graphs, evaluate $f(4)$.

a. 16

b. 18

c. 8

d. 20

32 [Intermediate] • Target: Amplitude, vertical shifts and dilations of trigonometric graphs

Find the positive solutions to the quadratic equation $x^2 - 10x + 24 = 0$.

a. $x = 4$ and $x = 7$

b. $x = 4$ and $x = 6$

c. $x = -4$ and $x = -6$

d. $x = 10$ and $x = 24$

33 [Intermediate] • Target: Period and phase shifts of trigonometric graphs

Find the positive solutions to the quadratic equation $x^2 - 5x + 4 = 0$.

a. $x = 1$ and $x = 5$

b. $x = 1$ and $x = 4$

c. $x = -1$ and $x = -4$

d. $x = 5$ and $x = 4$

34 [Intermediate] • Target: Period and phase shifts of trigonometric graphs

For the model $f(x) = 3x + 9$ in period and phase shifts of trigonometric graphs, evaluate $f(3)$.

a. 18

b. 21

c. 9

d. 22

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35 [Intermediate] • Target: Period and phase shifts of trigonometric graphs

Find the positive solutions to the quadratic equation $x^2 - 9x + 18 = 0$.

a. $x = 3$ and $x = 7$

b. $x = 3$ and $x = 6$

c. $x = -3$ and $x = -6$

d. $x = 9$ and $x = 18$

36 [Intermediate] • Target: Period and phase shifts of trigonometric graphs

Find the positive solutions to the quadratic equation $x^2 - 11x + 28 = 0$.

a. $x = 4$ and $x = 8$

b. $x = 4$ and $x = 7$

c. $x = -4$ and $x = -7$

d. $x = 11$ and $x = 28$

37 [Intermediate] • Target: Solutions to trigonometric equations

A data set recorded for solutions to trigonometric equations consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean?

a. 20.5

b. 21.5

c. 22.5

d. 24.5

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Solutions to trigonometric equations • Cross-Lesson: 3.01 Geometry & Trigonometry

For the model $f(x) = 4x + 3$ in solutions to trigonometric equations, evaluate $f(4)$.

a. 19

b. 23

c. 16

d. 23

39 [Challenging] • Target: Problem solving with trigonometric functions • Cross-Lesson: 4.01 Statistics & Probability

A data set recorded for problem solving with trigonometric functions consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean?

a. 21.5

b. 22.5

c. 23.5

d. 25.5

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40 [Challenging] • Target: Double angle identities • Cross-Lesson: 5.01 Calculus

A geometric figure scaled by factor $k = 2$ has an original side length of 8 cm. What is the corresponding image length?

a. 14 cm

b. 16 cm

c. 24 cm

d. 32 cm

41 [Challenging] • Target: Double angle identities • Cross-Lesson: 1.01 Number & Algebra

In ΔABC studying double angle identities, $m\angle A = 49^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 79°

b. 89°

c. 71°

d. 131°

42 [Challenging] • Target: Double angle identities • Cross-Lesson: 2.01 Functions

In ΔABC studying double angle identities, $m\angle A = 49^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 72°

b. 82°

c. 64°

d. 131°

43 [Challenging] • Target: Double angle identities • Cross-Lesson: 3.01 Geometry & Trigonometry

A geometric figure scaled by factor $k = 4$ has an original side length of 8 cm. What is the corresponding image length?

a. 28 cm

b. 32 cm

c. 40 cm

d. 128 cm

44 [Challenging] • Target: Double angle identities • Cross-Lesson: 4.01 Statistics & Probability

On the coordinate plane, a line with slope $m = 2$ passes through $(0, 5)$. What is its y-coordinate when $x = 4$?

a. 11

b. 15

c. 13

d. 15

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45 [Challenging] • Target: Linear regression line • Cross-Lesson: 5.01 Calculus

For the model $f(x) = 5x + 5$ in linear regression line, evaluate $f(3)$.

a. 20

b. 25

c. 15

d. 24

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